

AKSHAY B

Ernakulam, Kerala

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CAREER SUMMARY

Electronics and Communication Engineering student specializing in Embedded Systems, IoT, and AI-integrated applications. Experienced in building end-to-end solutions combining hardware (ESP32, sensors) and software (APIs, automation, AI tools). Developed real-world projects including intelligent automation tools, healthcare assistants, and computer vision systems. Currently expanding expertise in full-stack development and scalable AI-driven systems.

EDUCATION

Bachelor of Engineering in Electronics and Communication Cochin University of Science and Technology (CUSAT), Ernakulam	2023 – Present
Higher Secondary (Class XII) Sacred Heart Public School, Kerala	2020 – 2022
CCBP 4.0 Certification Program NxtWave, India	2023 – Present

SKILLS

Programming	Python, JavaScript (ES6+), C/C++
Web	HTML, CSS, Frontend Development
Embedded	ESP32, Arduino, Sensors, Embedded C
Databases	SQL, NoSQL
Tools	MATLAB, ANSYS, Make.com
AI/ML	TensorFlow (Basics), AI Agents, API Integration

EXPERIENCE

Embedded Systems Intern <i>Evolve Robotics, Ernakulam</i>	May 2025 – June 2025
· Worked on microcontroller-based system development and hardware interfacing. · Gained practical exposure to embedded programming and circuit integration.	

PROJECTS

ProjectETHack – AI-Powered Personalized News Platform

- Developed a role-based AI news generation system delivering personalized content based on user profession and interests.
- Built a TypeScript frontend and integrated Python backend for AI-driven content generation.
- Implemented API-based architecture for scalable and dynamic content delivery.
- **GitHub:** <https://github.com/zodiac0149/ProjectETHack>

MedCompanion – AI-Based Healthcare Assistant

- Designed an AI-powered healthcare assistant for chronic disease support including medication reminders and symptom tracking.
- Built a full-stack application using TypeScript (frontend) and Python (backend).
- Implemented AI-based health guidance system for user assistance.
- **GitHub:** <https://github.com/zodiac0149/ProjectGenAi2>

EpicTranslator – Sign Language Detection System

- Developed a computer vision-based system using OpenCV for real-time sign language recognition.
- Implemented multi-language translation with a web interface.
- Combined Python-based image processing with frontend technologies.
- **GitHub:** <https://github.com/zodiac0149/EpicTranslator>

ESP32-Based Rotatable Solar Panel System

- Designed a solar tracking system using ESP32 and LDR sensors to optimize sunlight exposure.
- Implemented embedded control logic for automated panel movement.
- Improved energy efficiency compared to static panel systems.

AI-Based ATS Resume Checker (Telegram Bot)

- Built an AI-powered resume analyzer using Telegram Bot API.
- Automated resume evaluation and feedback delivery via email using Make.com workflows.
- Enabled scalable and efficient resume review system.

CERTIFICATIONS

- Frontend Developer – NxtWave
- SQL Developer – NxtWave
- Python Developer – NxtWave

LANGUAGES

- English (Fluent)
- Malayalam (Native)
- Hindi (Basic)

HOBBIES & INTERESTS

- Exploring and experimenting with emerging AI tools and integrating them into web applications
- Learning and building with new technologies across IoT and software domains
- Reading tech blogs and staying updated with industry trends